EXPLOITING SPARSITY FOR PHASE UNWRAPPING

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ABSTRACT

We consider the problem of unwrapping the phase of two-dimensional interferograms, and adopt a known formulation as a sparse optimization problem. Many algorithms have been developed for solving sparse optimization problems that occur in the field of compressive sensing; in this work, we adapt one such algorithm for use in the unwrapping problem. The result is an unwrapping algorithm that gives very similar results to those of existing algorithms, but that is simpler, more reliable, and more computationally efficient.

1. INTRODUCTION

There are several applications that use the phase of an electromagnetic wave to derive useful information. These include interferometric synthetic aperture radar (InSAR) for detecting deformations of the Earth's surface, magnetic resonance imaging of the human body, interferometric measurements of mechanical properties of solid materials, and electron holography for measuring electromagnetic fields [1].

In these and other applications, the quantity of interest could be termed an *absolute* phase, which could be given in terms of a number of wavelengths. However, the process of measuring the phase only gives a *relative* phase, namely the fractional part of the number of wavelengths. The phase is often specified as an angle in radians, making 2π equivalent to one wavelength. Then relative phase corresponds to the measurement not being able to distinguish between a phase of 2π and a phase of 0 (or between a phase of θ and $\theta - 2\pi$, for any angle θ); the phase "wraps," just as adding 2π to an angle of the unit circle wraps around the circle to the same point. The measurement identifies a point on the circle, without knowing how many times the phase has gone around the circle. In order to extract scientific value from the measurement, we need to *unwrap* the phase, and determine values that are not limited to an interval of length 2π.

2. PHASE UNWRAPPING

2.1. Gradient adjustment

Many algorithms have been developed for unwrapping phase images; see [2] and the references therein for an overview. A common approach is to make use of gradient information from the wrapped phase; this gradient should differ from the gradient of the unwrapped phase only at pixels where wrapping has occurred. Moreover, the gradient itself can be an indicator of wrapping, as the discontinuity created by wrapping will tend to result in gradients of large magnitude.

To formalize this notion, consider differences between adjacent pixels, which we can model as discrete approximations of partial derivatives. So if our wrapped phase image is $\theta = \theta(i,j)$, define:

$$\begin{aligned} D_x\theta(i,j) &= \theta(i,j+1) - \theta(i,j), \\ D_y\theta(i,j) &= \theta(i+1,j) - \theta(i,j); \end{aligned} \tag{1}$$

in other words, we take the value at the next pixel minus the value at the current pixel. (We define $D_x\theta$ to be zero on the last column, and $D_y\theta$ to be zero on the last row.) Let's also assume that our wrapped phase values lie in the interval $(-\pi, \pi]$. Then our partial derivatives must lie in $(-2\pi, 2\pi)$. Since we may not know *a priori* where the wrapping boundary lies, we can use $|D_x\theta| > \pi$ or $|D_y\theta| > \pi$ as a heuristic for identifying wrapping, as this characterizes the property that $|D_x\theta|$ or $|D_y\theta|$ can be made smaller by adding 2π to or subtracting 2π from one of the phase values. This motivates the following (well-known) adjustment to the derivatives:

$$\varphi_x = \begin{cases} D_x\theta & \text{if } |D_x\theta| \le \pi, \\ D_x\theta - 2\pi\,\mathrm{sign}(D_x\theta) & \text{if } |D_x\theta| > \pi; \end{cases} \tag{2}$$

$$\varphi_y = \begin{cases} D_y\theta & \text{if } |D_y\theta| \le \pi, \\ D_y\theta - 2\pi\,\mathrm{sign}(D_y\theta) & \text{if } |D_y\theta| > \pi. \end{cases} \tag{3}$$

The result of this is to replace the gradient $D\theta = \binom{D_x\theta}{D_y\theta}$ with a vector field $\vec{\varphi} = \binom{\varphi_x}{\varphi_y}$, with the property that $|\varphi_x|$ and $|\varphi_y|$ are both at most π everywhere, with the expectation that $\vec{\varphi}$ will more closely match the gradient of the unwrapped phase.

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2.2. Integration

We seek a phase image Φ having $\vec{\varphi}$ as a gradient. However, it will often be the case the $\vec{\varphi}$ is not the gradient of any image. For a vector field to be a gradient is characterized by the property that the curl is everywhere zero, where the curl of a two-dimensional vector field $\vec{u} = \binom{u_x}{u_y}$ is defined by

$$\operatorname{curl} \vec{u} = D_x u_y - D_y u_x. \tag{4}$$

At least in the case of InSAR, it is common that $\vec{\varphi}$ will have nonzero curl at many pixels, often occurring in pairs of nearby pixels with curls of opposite sign, known as residues.

A natural approach, then, is to find the image whose gradient is as close as possible to $\vec{\varphi}$. This raises the issue of how to measure closeness; a simple way is to use a Euclidean (or ℓ^2) distance, giving us the following optimization problem [3]:

$$\min_{\Phi} \|D\Phi - \vec{\varphi}\|_2^2. \tag{5}$$

Solving (5) is equivalent to solving the following linear equation:

$$(D_x^T D_x + D_y^T D_y)\Phi = D_x^T \varphi_x + D_y^T \varphi_y. \tag{6}$$

If $\vec{\varphi}$ is curlfree, then the solution of (6) will have a gradient that is exactly $\vec{\varphi}$. However, in the more realistic case of $\vec{\varphi}$ having nonzero curl at some if not many pixels, the solution of (6) will typically have less dynamic range than the true, unwrapped phase, with the influence of residues having a global effect on the solution.

2.3. Sparse integration

An alternative that has often been proposed in the literature ([4, 5], for example) is to find Φ whose gradient differs from $\vec{\varphi}$ at as few pixels as possible. This amounts to solving

$$\min_{\Phi} \|D\Phi - \vec{\varphi}\|_0, \tag{7}$$

where the ℓ^0 norm $\|\cdot\|_0$ simply counts the number of nonzero pixels. Solving this problem directly is intractable, being NP-hard [5]. However, it is equivalent [6] to the type of problem considered in *compressive sensing* (CS; [7, 8]). This is because as Φ ranges over all scalar fields, $D\Phi$ ranges over all vector fields $\vec{\psi}$ whose curl is zero. Then by the substitution $\vec{\psi} = D\Phi - \vec{\varphi}$, (7) is equivalent to:

$$\min_{\vec{\psi}} \|\vec{\psi}\|_0 \text{ subject to } \operatorname{curl} \vec{\psi} = -\operatorname{curl} \vec{\varphi}. \tag{8}$$

This shows us that our problem is equivalent to the compressive sensing problem of finding the sparsest solution to a measurement problem. This suggests that algorithms used in compressive sensing may be successful at solving (7). The solution approach of CS is to replace the ℓ^0 norm with one that is easier to minimize, yet often gives the same solutions. The ℓ^1 norm is the one used most commonly, due to its convexity. However, the use of nonconvex penalty functions instead has been shown to be more effective [9, 10].

2.4. Splitting algorithm

We use the alternating directions, method of multipliers (ADMM) algorithm [11, 12] approach of [13], modified for the current problem. In particular we use the penalty function G_0 of [10], which was constructed to both promote sparsity and be efficient to minimize. We introduce a new variable $\vec{w}$, and penalize the difference between it and $D\Phi - \vec{\varphi}$, also introducing a Lagrange multiplier $\vec{\Lambda}$:

$$\min_{\Phi, \vec{w}} G_0(\vec{w}) + \tfrac{1}{2}\|\vec{w} - D\Phi + \vec{\varphi} - \vec{\Lambda}\|_2^2. \tag{9}$$

We now alternately solve for Φ and $\vec{w}$. At each iteration:

- Fix $\vec{w}$ and solve for Φ. This entails solving the following:

$$\begin{aligned}&(D_x^T D_x + D_y^T D_y)\Phi \\ &\quad = D_x^T(w_x + \varphi_x - \Lambda_x) + D_y^T(w_y + \varphi_y - \Lambda_y).\end{aligned} \tag{10}$$

  The derivative operators D_x, D_y are diagonalized by the discrete cosine transform, which allows us to solve (10) using fast cosine transforms.

- Fix Φ and solve for $\vec{w}$, which by construction of G_0 is given by the formula

$$\vec{w} = \max\{|\vec{x}| - 1/|\vec{x}|, 0\} \operatorname{sign}(\vec{x}), \tag{11}$$

  with $\vec{x} = D\Phi - \vec{\varphi} + \vec{\Lambda}$, and where operations are done pixelwise, with the result taken to be 0 where $\vec{x} = 0$.

- The Lagrange multiplier $\vec{\Lambda}$ is updated by adding the residual $D\Phi - \vec{\varphi} - \vec{w}$. This uses the method of multipliers [14, 15] to force $\vec{w}$ and $D\Phi - \vec{\varphi}$ to converge together.

The algorithm is computationally very efficient, with the fast cosine transforms dominating the computational cost.

3. INSAR EXAMPLE

We consider an interferogram computed from a pair of single-look-complex (SLC) images from the ESA SAR satellite Sentinel 1-A, of a region including a portion of Mexico City, taken on September 25 and November 12, 2016. The wrapped phase and coherence, each 12188×5287 pixels (in range-azimuth coordinates), are in Fig. 1. We compare our unwrapping result with that of SNAPHU [5], a commonly-used phase unwrapping algorithm; it is distributed by the European Space Agency (ESA) for use with their Sentinel Application Platform (SNAP; `http://step.esa.int/main/toolboxes/snap/`), for use in processing InSAR imagery from Sentinel-1. The approach of [5] is based upon a network-optimization approach

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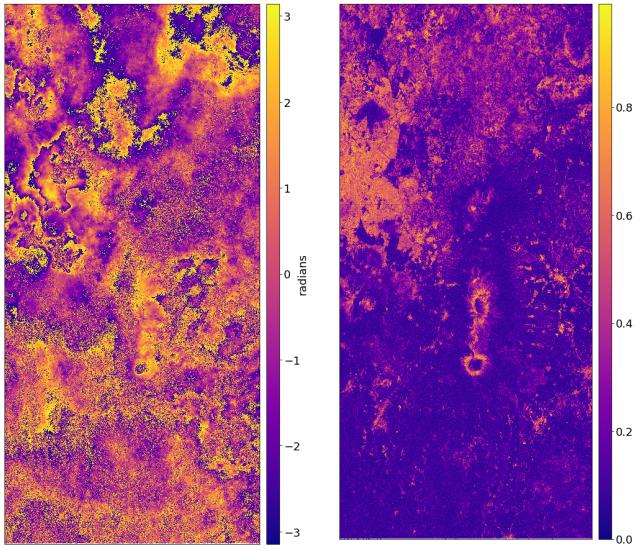

Fig. 1. Left: wrapped interferogram phase for an interferogram in the Mexico City vicinity. Right: interferogram coherence.

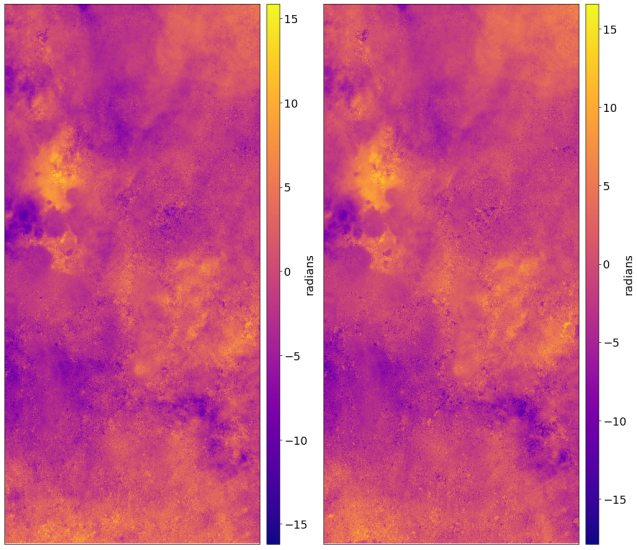

Fig. 2. Left: unwrapped phase using our algorithm. Right: Unwrapped phase using SNAPHU. The results are nearly the same.

to solving (7), though with additional application-specific customizations that we do not consider.

The algorithm described in Sec. 2.4 was run for 10 iterations, with our Python implementation running in 565 seconds on a Google Cloud Platform virtual machine (VM) with 32 virtual CPUs and 120 GB of RAM. The SNAPHU algorithm, implemented in C in the GMT5SAR library [16], was run on the same VM, taking 4558 seconds, with parameters heavily optimized for computational efficiency. The results are in Fig. 2 and Fig. 3. We find that the results are nearly identical, so that our result is of the same quality, but obtained in a small fraction of the time.

We also find our algorithm to give much more reliable results. We ran our algorithm and SNAPHU on 100 interferograms of the same region but using pairs of SLC images collected at various times and with the timespan between the two images varying between 12 and 744 days. We found that in 73 of the cases, SNAPHU failed to converge in the four hours we allotted. In the case of our algorithm, all unwrappings took between 475 and 581 seconds and produced meaningful output. See Fig. 4 for results from our algorithm in the difficult case of an SLC pair from November 18, 2015 and December 1, 2017, where SNAPHU failed to produce a result.

4. CONCLUSION

In this paper we consider a known formulation of the phase unwrapping problem in terms of sparsity, that of finding the phase image that differs from gradient information from the wrapped phase at as few pixels as possible. This motivates applying algorithm methods from the field of compressive sensing, which has seen an enormous amount of research in finding the sparsest solutions to linear inverse problems. We adapt an algorithm which was very successful in the context of compressive sensing, in particular solving a nonconvex approximation of the original problem. The result is a very efficient algorithm, which in examples from InSAR gives unwrapping results nearly identical to that of SNAPHU, a commonly-used unwrapping algorithm for InSAR, in s small fraction of the time and with greater reliability.

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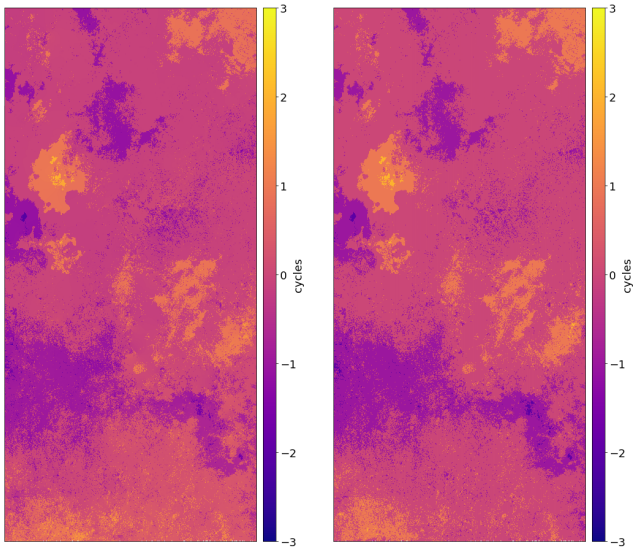

Fig. 3. Difference between the unwrapped and wrapped phase, using our algorithm (left) and SNAPHU (right). The step-function nature of the (nearly identical) residuals, particularly in regions of high coherence, is as desired.

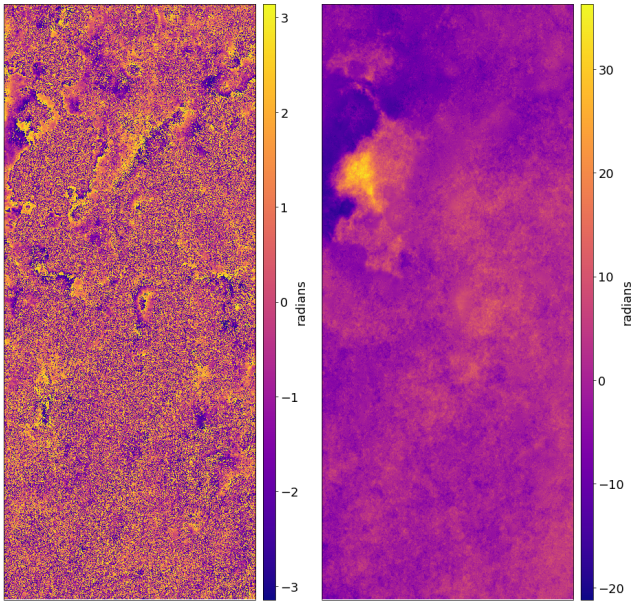

Fig. 4. Left: wrapped phase of an interferogram using SLC images collected 744 days apart. Right: unwrapped phase using our algorithm is reasonable, with computation time under 10 minutes; SNAPHU failed to return a result in 12 hours.

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